MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

THIRD SEMESTER EXAMINATION

(2024 ADMISION ONWARDS)

MG3DSEECT201 3D PRINTING AND ITS APPLICATIONS

Duration: 2 hours **Maximum Marks: 70**

*Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

Part A

Multiple Choice Questions

Answer All Questions, each question carries 1 mark

1. Additive Manufacturing is best described as a process of:
   a) Removing material layer by layer b) Joining materials by welding
   c) Creating objects by adding material layer by layer
   d) Casting molten material into molds **U 1**
2. Which software is commonly used for 3D modeling in additive manufacturing?
   a) MS Excel b) AutoCAD / Fusion 360
   c) MATLAB d) SPSS **U 2**
3. The main function of the extruder in a 3D printer is to:
   a) Cool the printed part b) Feed and melt the filament
   c) Move the build platform d) Control power supply **An 3**
4. Design for Manufacturability (DfAM) in additive manufacturing mainly aims to:
   a) Increase design complexity without limits
   b) Ensure designs can be efficiently and reliably 3D printed
   c) Replace CAD software d) Increase post-processing steps **C 4**
5. Which of the following is an extrusion-based 3D printing process?
   a) Stereolithography (SLA) b) Selective Laser Sintering (SLS)
   c) Fused Deposition Modeling (FDM) d) Binder Jetting **U 1**
6. In computational design of a key chain, constraints are used to:
   a) Increase material wastage b) Control size, shape, and functional requirements
   c) Eliminate design rules d) Reduce design flexibility **U 2**
7. Personalization in AM is best suited for:
   a) Large-scale casting b) One-size-fits-all products
   c) Customized products like name tags or keychains
   d) Heavy industrial forging **An 3**
8. Support structures in 3D printing are mainly used to:
   a) Improve material strength b) Support overhangs and complex geometries
   c) Increase print speed d) Change filament color **C 4**
9. A major advantage of Binder Jetting is:
   a) No need for support structures b) High surface finish without post-processing
   c) Very high strength parts directly d) Only polymers can be used **U 1**
10. Generative design in AM is best described as:
    a) Copying existing designs b) Creating designs using predefined shapes
    c) Using algorithms to generate optimized design solutions
    d) Designing without software **U 2**
11. Stepper motors in 3D printers are mainly used for:

a) Heating the filament b) Accurate positioning and movement
c) Cooling the hot end d) Storing print files **An 3**

12. Which design feature MOST often requires support material?
a) Vertical walls b) Solid cubes
c) Overhangs and bridges d) Flat bases **C 4**

13. Selective Laser Sintering (SLS) belongs to which category?
a) Extrusion-based printing b) Vat photopolymerization
c) Powder-based printing d) Droplet-based printing **U 1**

14. Slicing software in AM is used to:
a) Convert 2D drawings into 3D models b) Cut physical materials
c) Convert 3D models into layer-wise instructions
d) Test mechanical properties **U 2**

15. Bed levelling is a part of 3D printer calibration used to:
a) Increase print speed b) Ensure proper first-layer adhesion
c) Change filament colour d) Reduce power consumption **An 3**

16. Hollow structures in AM designs are preferred because they:
a) Increase weight b) Save material and reduce printing time
c) Eliminate calibration d) Reduce design flexibility **C 4**

1*16=16

Part B

Short Answer Questions
Answer any 6 questions, each question carries 5marks

17. Describe the working principle of extrusion-based 3D printing with a neat sketch. **U 1**
18. Explain the concept of parametric modelling with a simple example. **U 2**
19. Examine the main components of a 3D printer and state the function of any four. **An 3**
20. Compose design for Manufacturability (DfAM) in additive manufacturing. **C 4**
21. Differentiate between Material Jetting and Binder Jetting processes. **U 1**
22. Summarize any four principles of computational design for AM. **U 2**
23. Outline the difference between customization and personalization in 3D printing. **An 3**
24. Formulate the role of nozzle diameter in determining wall thickness. **C 4**

5*6=30

Part C

Essay Questions
Answer any 2 Questions, each questions carry 12 marks

25. Explain the classification of additive manufacturing processes. Describe extrusion-based, vat photopolymerization, and powder-based printing methods with suitable examples. **U 1**
26. Discuss the principles of computational design for additive manufacturing, highlighting parametric and generative approaches. **U 2**
27. Illustrate the importance of 3D printer calibration. Explain various calibration techniques used in practice. **An 3**
28. Compose support structures in 3D printing. Discuss their types, necessity, advantages, limitations, and removal methods. **C 4**

2*12=24